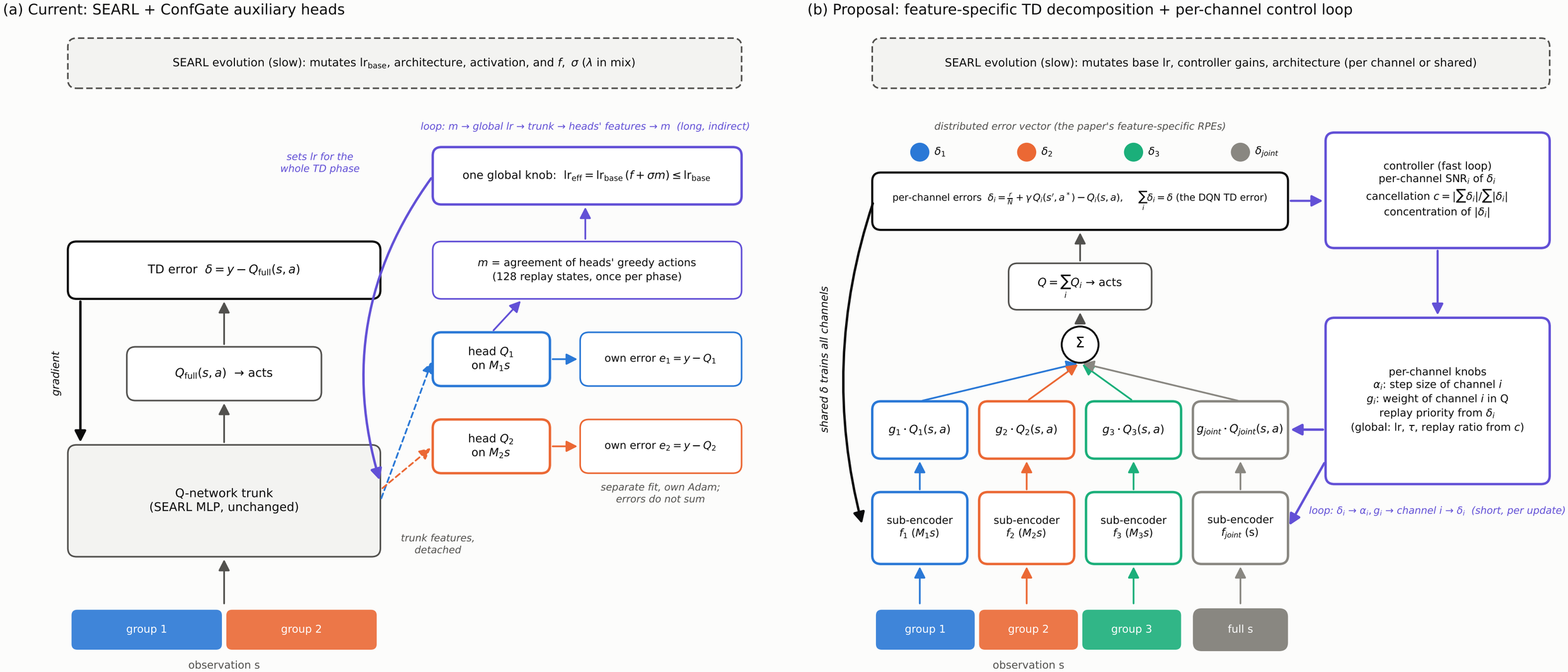
(a) Current: SEARL + ConfGate auxiliary heads
SEARL evolution (slow): mutates lr_{base}, architecture, activation, and f, σ (λ in mix)
loop: m → global lr → trunk → heads' features → m (long, indirect)
sets lr for the whole TD phase
one global knob: $lr_{eff} = lr_{base}(f + \sigma m) \le lr_{base}$
m = agreement of heads' greedy actions (128 replay states, once per phase)
TD error $\delta = y - Q_{full}(s,a)$
$Q_{full}(s,a)$ → acts
gradient
head Q_1 on $M_1 s$
own error $e_1 = y - Q_1$
head Q_2 on $M_2 s$
own error $e_2 = y - Q_2$
separate fit, own Adam; errors do not sum
Q-network trunk (SEARL MLP, unchanged)
trunk features, detached
group 1
group 2
observation s
(b) Proposal: feature-specific TD decomposition + per-channel control loop
SEARL evolution (slow): mutates base lr, controller gains, architecture (per channel or shared)
distributed error vector (the paper's feature-specific RPEs)
δ_1
δ_2
δ_3
δ_{joint}
per-channel errors $\delta_i = \frac{r}{N} + \gamma Q_i(s', a^*) - Q_i(s,a)$, $\sum_i \delta_i = \delta$ (the DQN TD error)
controller (fast loop)
per-channel SNR_i of δ_i
cancellation $c = |\sum \delta_i| / \sum |\delta_i|$
concentration of $|\delta_i|$
$Q = \sum_i Q_i$ → acts
Σ
shared δ trains all channels
per-channel knobs
α_i: step size of channel i
g_i: weight of channel i in Q
replay priority from δ_i
(global: lr, τ, replay ratio from c)
$g_1 \cdot Q_1(s,a)$
$g_2 \cdot Q_2(s,a)$
$g_3 \cdot Q_3(s,a)$
$g_{joint} \cdot Q_{joint}(s,a)$
loop: $\delta_i \to \alpha_i, g_i$ → channel i → δ_i (short, per update)
sub-encoder f_1 ($M_1 s$)
sub-encoder f_2 ($M_2 s$)
sub-encoder f_3 ($M_3 s$)
sub-encoder f_{joint} (s)
group 1
group 2
group 3
full s
observation s
Legend: black = shared error / gradient violet = fast control loop (within lifetime) dashed grey = evolution (across generations) coloured = observation groups / channels (and, in (c), their own errors)